

EduTrack LMS

Project Requirements & Task Breakdown

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Overview

EduTrack is a learning management system being built for mid-sized coaching institutes to manage course content, track student engagement, and automate deadline reminders. This document defines the task breakdown across three core modules: Content Management, User Analytics, and the Notification System. Each task includes an owner, deadline, and priority level for sprint planning.

Chapter 1: Content Management

This module handles course creation, video content delivery, and version control for instructors updating existing material.

TASK 1.1.1: Course Upload Interface

Owner: Ananya R. | Deadline: Sprint 4 (Apr 12) | Priority: High

Build a drag-and-drop interface allowing instructors to upload course materials including PDFs, slide decks, and video files up to 2GB per file. Uploaded files must be virus-scanned before being made available to students.

TASK 1.1.2: Video Transcoding Pipeline

Owner: Rohit K. | Deadline: Sprint 5 (Apr 26) | Priority: High

Implement an automated pipeline that transcodes uploaded videos into multiple resolutions (360p, 720p, 1080p) using FFmpeg, and generates thumbnail previews at the 10-second mark of each video.

TASK 1.2.1: Content Versioning

Owner: Ananya R. | Deadline: Sprint 6 (May 10) | Priority: Medium

Allow instructors to publish a new version of a lesson without deleting the old one. Students already partway through the old version should be notified of the update but not force-migrated mid-course.

Chapter 2: User Analytics Dashboard

Provides instructors and institute administrators visibility into student engagement, so that at-risk students can be identified early in the term.

TASK 2.1.1: Engagement Metrics

Owner: Priya M. | Deadline: Sprint 5 (Apr 26) | Priority: High

Track and display per-student metrics including video watch time, quiz attempt count, and days-since-last-login, updated on a rolling 24-hour basis.

TASK 2.1.2: Cohort Comparison Reports

Owner: Priya M. | Deadline: Sprint 7 (May 24) | Priority: Medium

Generate a weekly report comparing the current cohort's average quiz scores and engagement against the historical average of the same course from previous terms.

TASK 2.2.1: Data Export (CSV/PDF)

Owner: Vikram S. | Deadline: Sprint 7 (May 24) | Priority: Low

Allow administrators to export any analytics view as CSV for further analysis in spreadsheet tools, or as a formatted PDF suitable for sharing with institute management.

Chapter 3: Notification System

Keeps students informed of upcoming deadlines and new content across email, SMS, and in-app channels, with configurable preferences per student.

TASK 3.1.1: Email Digest Scheduler

Owner: Karan D. | Deadline: Sprint 8 (Jun 7) | Priority: Medium

Send a daily digest email at 7 AM local time summarizing assignments due within the next 48 hours, using a queued job system to handle institutes with over 10,000 students.

TASK 3.1.2: SMS Alerts for Deadlines

Owner: Karan D. | Deadline: Sprint 8 (Jun 7) | Priority: Medium

Send an SMS reminder 2 hours before an assignment deadline, only for assignments marked as high-priority by the instructor. Integrate with a third-party SMS gateway supporting Indian carrier numbers.

TASK 3.2.1: In-App Notification Center

Owner: Meera T. | Deadline: Sprint 9 (Jun 21) | Priority: High

Build a bell-icon notification center within the student dashboard showing the last 30 days of notifications, with read/unread status and the ability to jump directly to the related course content.

TASK 3.2.2: Push Notification Service

Owner: Meera T. | Deadline: Sprint 9 (Jun 21) | Priority: High

Implement mobile push notifications for the EduTrack Android and iOS apps, triggered by the same events as the email digest, using Firebase Cloud Messaging. Push notifications must respect the student's configured quiet hours.

Appendix A: Glossary

Term	Definition
Cohort	A group of students enrolled in the same course during the same term
Digest	A single summary notification bundling multiple individual updates
Quiet hours	A student-configured time window during which push notifications are suppressed